Model/
type
Serial
number
EGC System Unit and Technical
Manual approval number

A copy of this Certificate, together with the EGC System Technical Manual, shall be carried on board the ship fitted with this EGC System unit at all times.

This Certificate is valid for the life of the EGC System unit, subject to surveys in accordance with section 4.2 of the guidelines and regulation 5 of MARPOL Annex VI, installed in ships under the authority of this Government.

Issued at .....
(place of issue of certificate)

dd/mm/yyyy
....
(date of issue) (signature of duly authorized official
issuing the certificate)

(Seal or Stamp of the authority, as appropriate)
Badge
or
Cipher
RESOLUTION MEPC.259(68)
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APPENDIX 2

PROOF OF THE SO2/CO2 RATIO METHOD

1 The SO 2/CO2 ratio method enables direct monitoring of exhaust gas emissions to verify compliance with emissions limits set out in table 1 in paragraph 1.3 of these guidelines. In the case of EGC systems that absorb CO 2 during the exhaust gas cleaning process it is necessary to measure the CO 2 prior to the cleaning process and use the CO 2 concentration before cleaning with the SO 2 concentration after cleaning. For conventional low alkali cleaning systems virtually no CO 2 is absorbed during exhaust gas cleaning and therefore monitoring of both gases can be undertaken after the cleaning process.

2 Correspondence between the SO 2/CO2 ratio can be determined by simple inspection of the respective carbon contents per unit mass of distillate and residual fuel. For

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monitoring of both gases can be undertaken after the cleaning process.

2 Correspondence between the SO 2/CO2 ratio can be determined by simple inspection of the respective carbon contents per unit mass of distillate and residual fuel. For this group of hydrocarbon fuels the carbon content as a percentage of mass remains closely similar, whereas the hydrogen content differs. Thus it can be concluded that for a given carbon consumption by combustion there will be a consumption of sulphur in proportion to the sulphur content of the fuel, or in other words a constant ratio between carbon and sulphur adjusted for the molecular weight of oxygen from combustion.

3 The first development of the SO 2/CO2 ratio considered its use to verify compliance with emissions from 1.5% sulphur fuel. The limit of 65 (ppm 4/%) SO2/CO2 for 1.5% sulphur in fuel can be demonstrated by first calculating the mass ratio of fuel sulphur to fuel carbon, which is tabulated in table 1 in this appendix for various fuels and fuel sulphur contents; including 1.5% sulphur for both distillate and residual fuels. These ratios were used to solve

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which is tabulated in table 1 in this appendix for various fuels and fuel sulphur contents; including 1.5% sulphur for both distillate and residual fuels. These ratios were used to solve for the corresponding SO2 and CO2 concentrations in exhaust, which are tabulated in table 2 of this appendix. Molecular weights (MW) were taken into account to convert mass fractions to mole fractions. For the 1.5% sulphur fuels in table 2, the amount of CO 2 is set first at 8% and then changed to 0.5% to show that there is no effect due to changes in excess air. As expected, the absolute SO 2 concentration changes, but the SO 2/CO2 ratio does not. This indicates that the SO 2/CO2 ratio is independent of fuel- to-air ratios. Therefore, SO 2/CO2 ratio can be used robustly at any point of operation, including operation where no brake power is produced.

3.1 Note that the SO 2/CO2 ratio varies slightly from distillate to residual fuel. This occurs because of the very different atomic hydrogen- to-carbon ratios (H:C) of the two fuels. Figure 1 illustrates the extent of the SO 2/CO2 ratios' sensitivity to H:C over a broad range of

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because of the very different atomic hydrogen- to-carbon ratios (H:C) of the two fuels. Figure 1 illustrates the extent of the SO 2/CO2 ratios' sensitivity to H:C over a broad range of H:C and fuel sulphur concentrations. From Figure 1, it can be concluded that for fuel sulphur levels less than 3.0% sulphur, the difference in S/C ratios for distillate and residual fuel is less than 5.0%.

3.2 In the case of using non-petroleum fuel oils, the appropriate SO 2/CO2 ratio applicable to the values given in regulations 14.1 and/or 14.4 of MARPOL Annex VI will be subject to approval by the Administration.

4 ppm means "parts per million". It is assumed that ppm is measured by gas analysers on a molar basis, assuming ideal gas behaviour. The technically correct units are actually micro-moles of substance per mole of total amount (mol/mol), but ppm is used in order to be consistent with units in the NO x Technical Code.
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https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
Table 1: Fuel properties for marine distillate and residual fuel*

Fuel Type	Carbon %(m/m)	Hydrogen %(m/m)	Sulphur %(m/m)	Other %(m/m)	C mol/kg	H mol/kg	S mol/kg	Fuel mol/mol	S/C ppm/%(v/v)	Exh SO2/CO2
Distillate	86.20	13.60	0.17	0.03	71.8333	136	0.0531	0.00074	7.39559	
Residual	86.10	10.90	2.70	0.30	71.7500	109	0.8438	0.01176	117.5958	
Distillate	85.05	13.42	1.50	0.03	70.8750	134.2	0.4688	0.006614	66.1376	
Residual	87.17	11.03	1.50	0.30	72.6417	110.3	0.4688	0.006453	64.5291	

* Based on properties in the IMO NOx Monitoring Guidelines, resolution MEPC.103(49).

Table 2: Emissions calculations corresponding to 1.5 % fuel sulphur

	CO2 %	SO2 Exh ppm4	SO2/CO2 Exh ppm4/% m/m	S/C
Distillate 0.17% S	8	59.1	7.4	0.00197
Residual 2.70% S	8	939.7	117.5	0.03136
Distillate 1.5% S	8	528.5	66.1	0.01764

Residual 1.5% S 8 515.7 64.5 0.01721

Distillate 1.5% S 0.5 33.0 66.1 0.01764
Residual 1.5% S 0.5 32.2 64.5 0.01721

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Residual 2.70% S 8 939.7 117.5 0.03136

Distillate 1.5% S 8 528.5 66.1 0.01764
Residual 1.5% S 8 515.7 64.5 0.01721

Distillate 1.5% S 0.5 33.0 66.1 0.01764
Residual 1.5% S 0.5 32.2 64.5 0.01721

SO2/CO2 ratio vs % sulphur in fuel
0
50
100
150
200